

LEC25 - Series Continued

Warm-Up Problem

The graph shows the beginning of a sequence of partial sum S_n . Which of the options could be the corresponding sequence $\{a_n\}$?

- a) $a_n = \frac{2}{n}$ b) $a_n = (-1)^n \cdot \frac{1}{n}$ c) $a_n = n^2$ d) $a_n = (-\frac{5}{7})^n$ e) $a_n = (-\frac{2}{3})^n$

⇒ Some things to note:

- S_n is divergent
- S_n is alternating

Example

The Earth is estimated to contain 140 tons of a certain non-renewable resource. Currently, 12 tons are consumed each year worldwide. Suppose that this year we consume 12 tons, but starting next year we are able to decrease the consumption by $x\%$ each year. What would x need to be in order for the resource to never run out? Or is that not possible no matter what x is?

$$\begin{aligned} \Rightarrow a_n &= 12(1-x)^n \\ \Rightarrow \sum_{n=1}^{\infty} 12(1-x)^n &= 140 \quad \text{This is a geometric series, which we know must converge.} \\ \Rightarrow \sum_{n=1}^{\infty} (1-x)^n &= \frac{140}{12} \quad \text{Therefore, the series must converge to } \frac{1}{1-r}, \text{ where } r=1-x \\ \Rightarrow \frac{1}{1-x} &= \frac{140}{12} \Rightarrow \frac{1}{1-x} = \frac{140}{12} \\ \Rightarrow x &= \frac{12}{140} \approx 8.57\% \end{aligned}$$

Example

Find the sum of this series, if it converges: $\sum_{n=3}^{\infty} \frac{2}{n(n-2)}$

⇒ Rewrite using partial fractions

$$\rightarrow \frac{2}{n(n-2)} = \frac{A}{n} + \frac{B}{n-2}$$

$$\rightarrow 2 = A(n-2) + B(n)$$

$$\bullet \text{ Let } n=2 \Rightarrow 2 = 2B \Rightarrow 1 = B$$

$$\bullet \text{ Let } n=0 \Rightarrow 2 = A(-2) \Rightarrow -1 = A$$

$$\left. \begin{aligned} \bullet \text{ Let } n=2 \Rightarrow 2 = 2B \Rightarrow 1 = B \\ \bullet \text{ Let } n=0 \Rightarrow 2 = A(-2) \Rightarrow -1 = A \end{aligned} \right\} \frac{2}{n(n-2)} = \frac{1}{n-2} - \frac{1}{n}$$

⇒ Write out the first few terms

$$S_n = 1 - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \frac{1}{4} - \frac{1}{6} \quad \text{These terms will cancel if more terms are written out}$$

⇒ Since the series is telescoping, all terms will cancel out except for $1 + \frac{1}{2} = \frac{3}{2}$

⇒ Therefore, the series converges to $\frac{3}{2}$

Example

Suppose $\sum_{n=1}^{\infty} a_n$ is a series whose sequence of partial sums is $S_n = 2 - \frac{10}{3n}$.

1. What is $\sum_{n=1}^{\infty} a_n$?

$$= S_{10} = 2 - \frac{10}{3(10)} = 2 - \frac{1}{3} = \frac{5}{3}$$

2. Does the series converge, and if so, what is its sum?

$$\lim_{k \rightarrow \infty} S_k = \lim_{k \rightarrow \infty} \left[2 - \frac{10}{3k} \right] = 2, \text{ therefore the sequence converges to } 2$$

3. What is a_{10} ?

$$\Rightarrow S_n - S_9 = a_{10} = 2 - \frac{10}{3n} - \left(2 - \frac{10}{3(9)} \right) = a_{10} = 2 - \frac{10}{3(10)} - 2 + \frac{10}{3(9)} = \frac{1}{11}$$

4. What is a_n ?

$$\Rightarrow S_n - S_{n-1} = a_n = 2 - \frac{10}{3n} - \left(2 - \frac{10}{3(n-1)} \right) = a_n = 2 - \frac{10}{3n} - 2 + \frac{10}{3(n-1)} = \frac{10}{3n-3} - \frac{10}{3n}$$